

Rollno:241001051

SOLVE A GRAPH USING DEPTH FIRST SEARCH(DFS)

Python code:

```
defdfs(graph, start, visited=None):
```

```
    ifvisited is None:
```

```
        visited = set()
```

```
    visited.add(start)
```

```
    print(start, end=" ")
```

```
    forneighbor in graph[start]:
```

```
        ifneighbor not in visited:
```

```
            dfs(graph, neighbor, visited)
```

```
graph={
```

```
    'A':['B', 'C'],
```

```
    'B':['D', 'E'],
```

```
    'C':['F'],
```

```
'D': [],  
'E': ['F'],  
'F': []  
}
```

```
print("DFS traversal from node A:")  
dfs(graph, 'A')
```

output:

DFS traversal from node A:

A B D E F C

Result:

Thus the program of implementing DFS traversal on a graph to explore all nodes was executed successfully.